Zifan Si - SVM Solar Flare Classification: Report Outputs

[1/2] FEATURE EXPERIMENT (2010–2015) — finding best feature set

Best feature-set COMBO on 2010–2015: FS-I+FS-IV (TSS mean ± std = 0.7773 ± 0.0395).

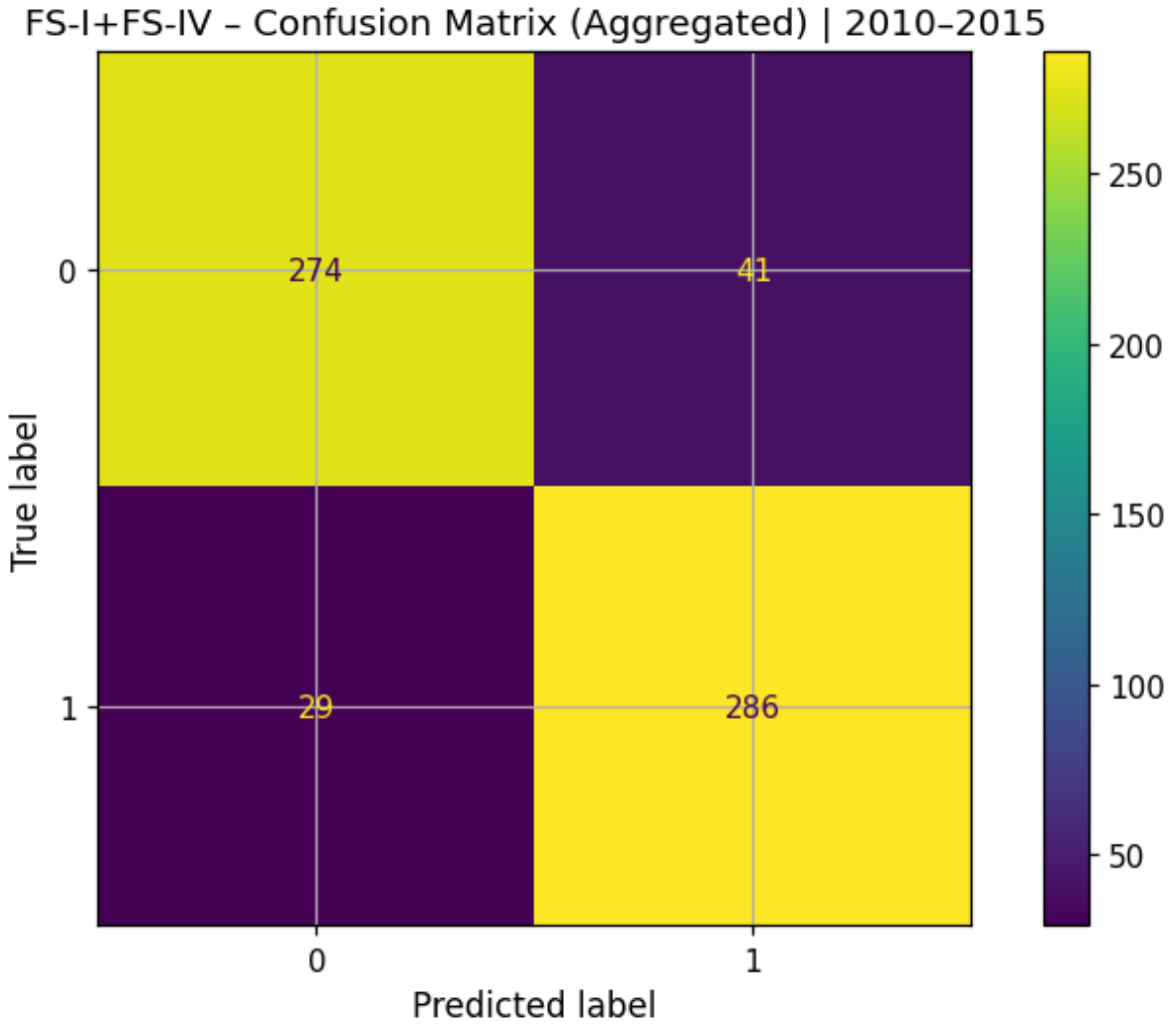


Figure 1: Feature Experiment — Plot 1

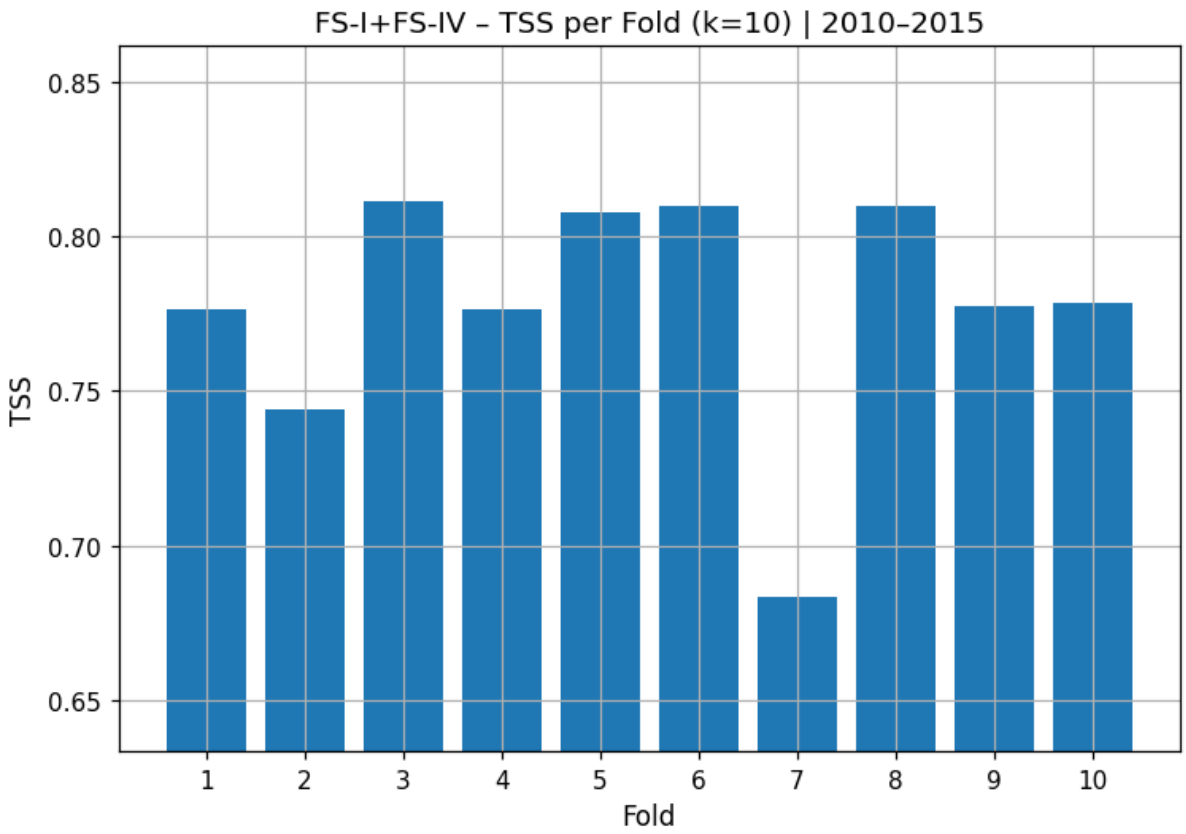


Figure 2: Feature Experiment — Plot 2

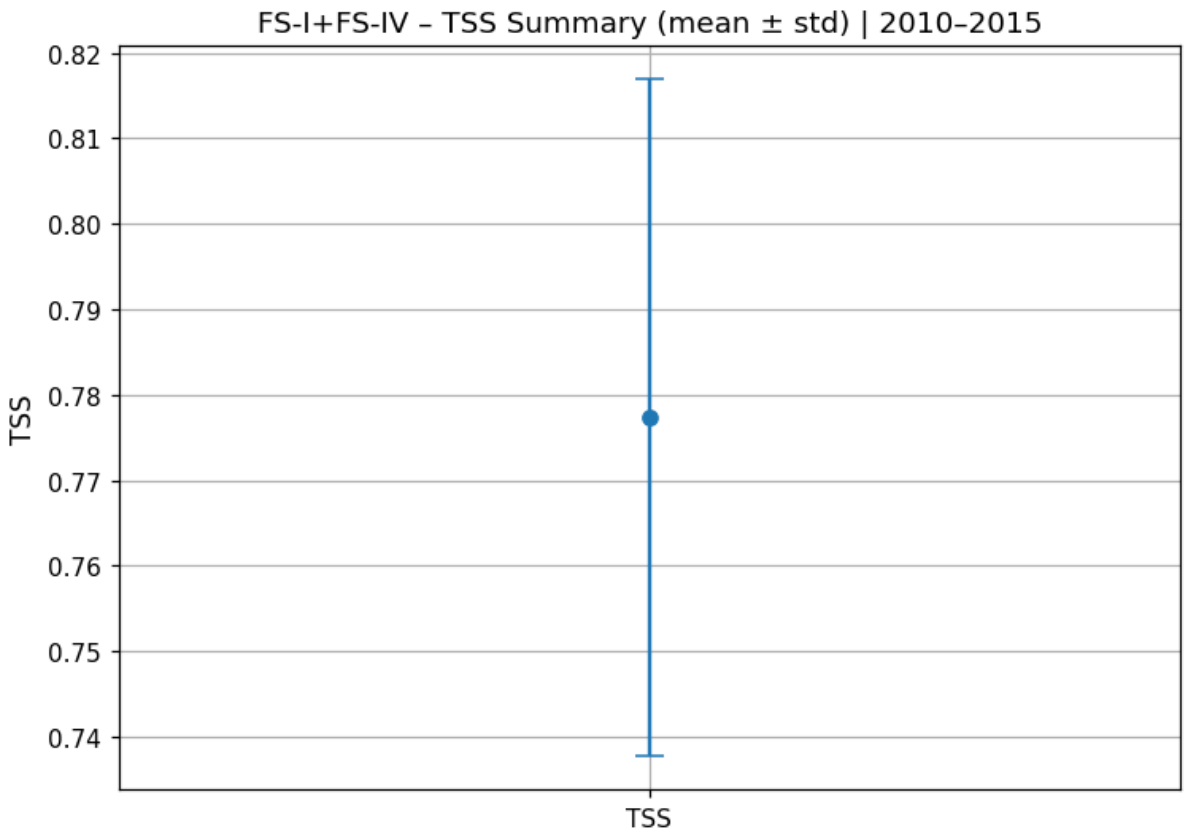


Figure 3: Feature Experiment — Plot 3

[2/2] DATA EXPERIMENT (2010–2015 & 2020–2024) — evaluating best feature set

Using best feature-set from 2010–2015: FS-I+FS-IV

2010–2015

data-2010-15 [FS-I+FS-IV]: TSS mean ± std = 0.7773 ± 0.0395 (k=10).

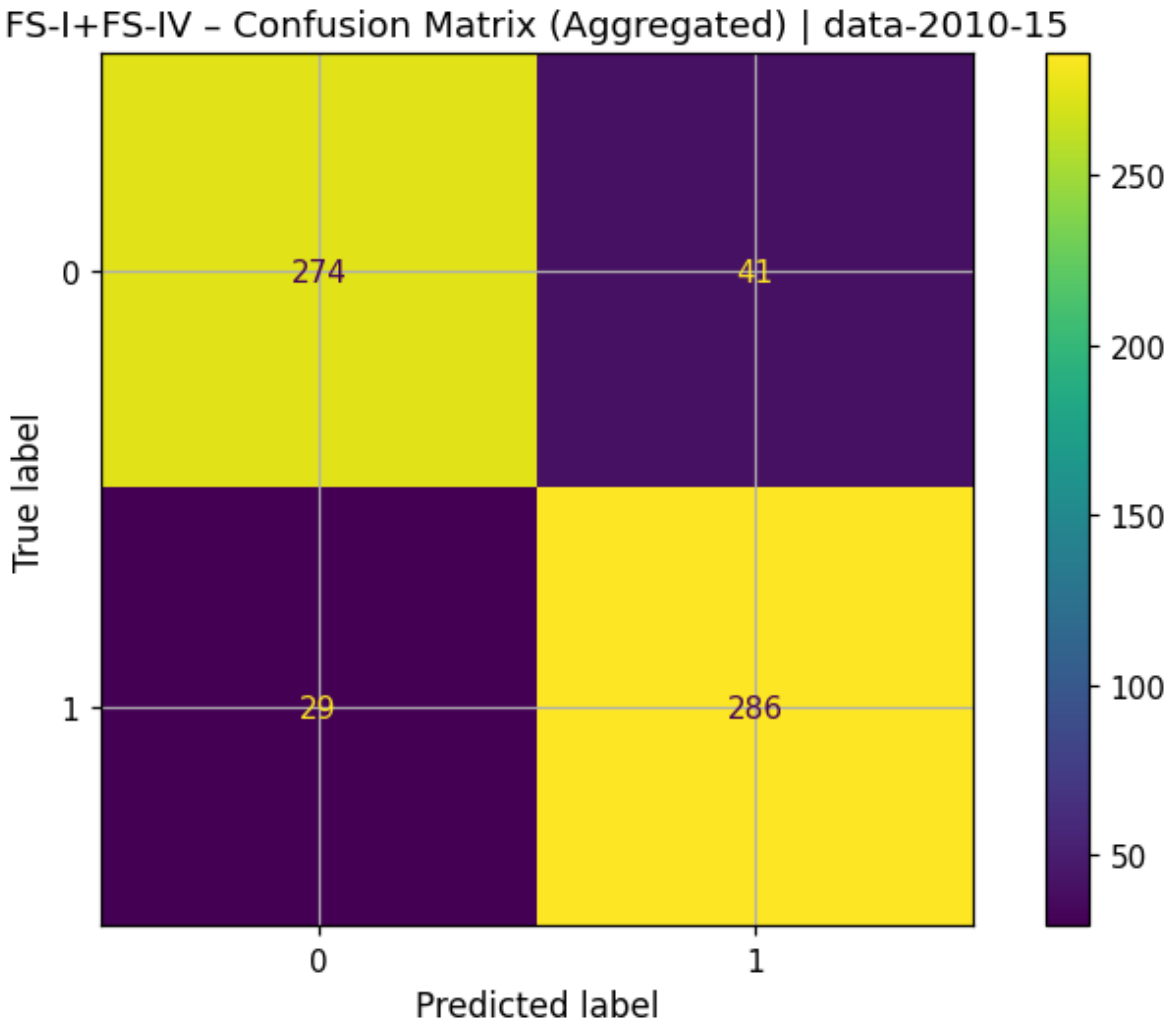


Figure 4: 2010–2015 — Confusion Matrix

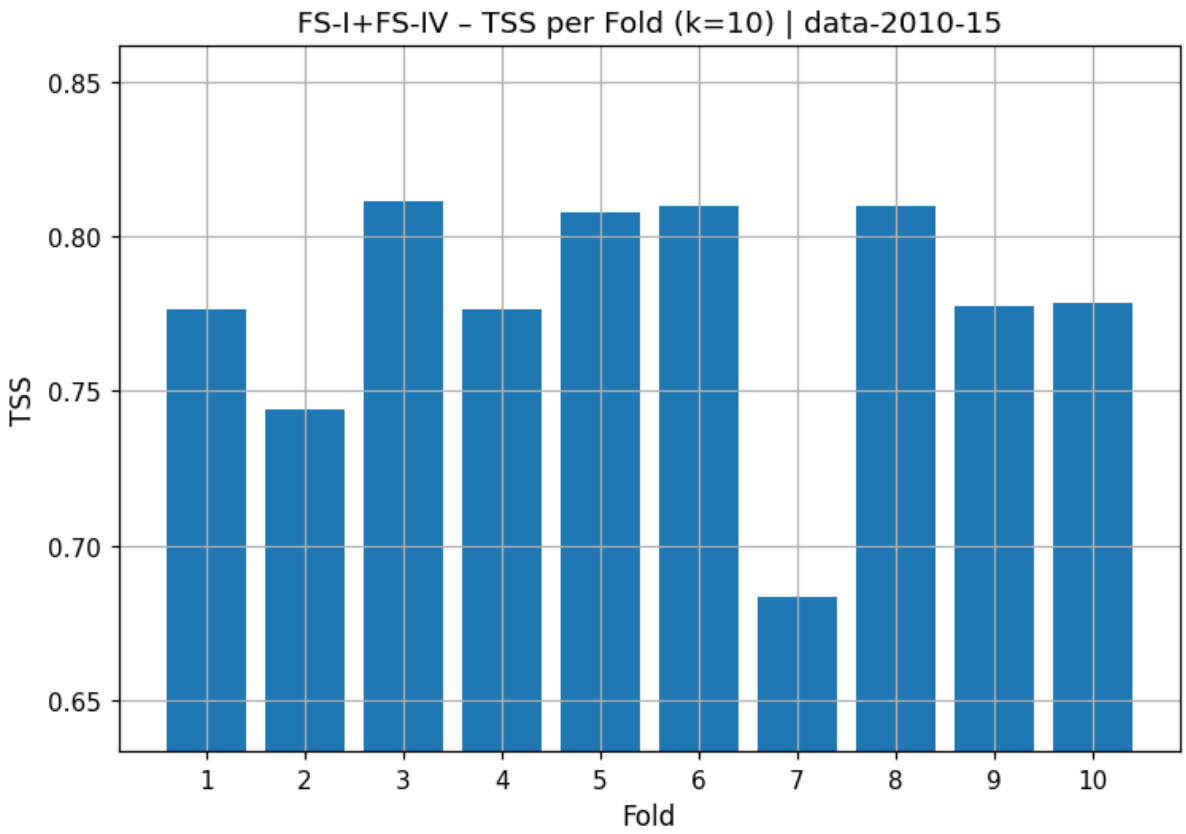


Figure 5: 2010–2015 — TSS per Fold

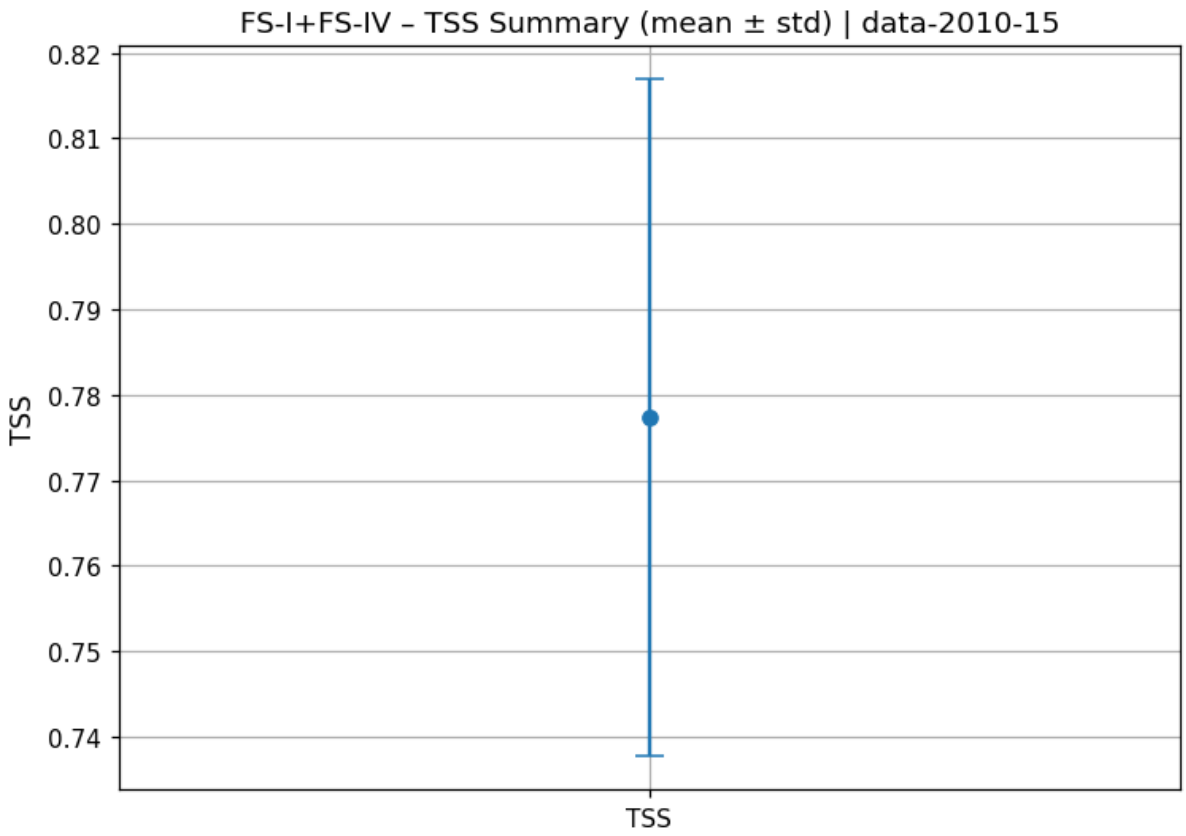


Figure 6: 2010–2015 — TSS Summary (mean ± std)

2020–2024

data-2020-24 [FS-I+FS-IV]: TSS mean ± std = 0.6619 ± 0.2173 (k=10).

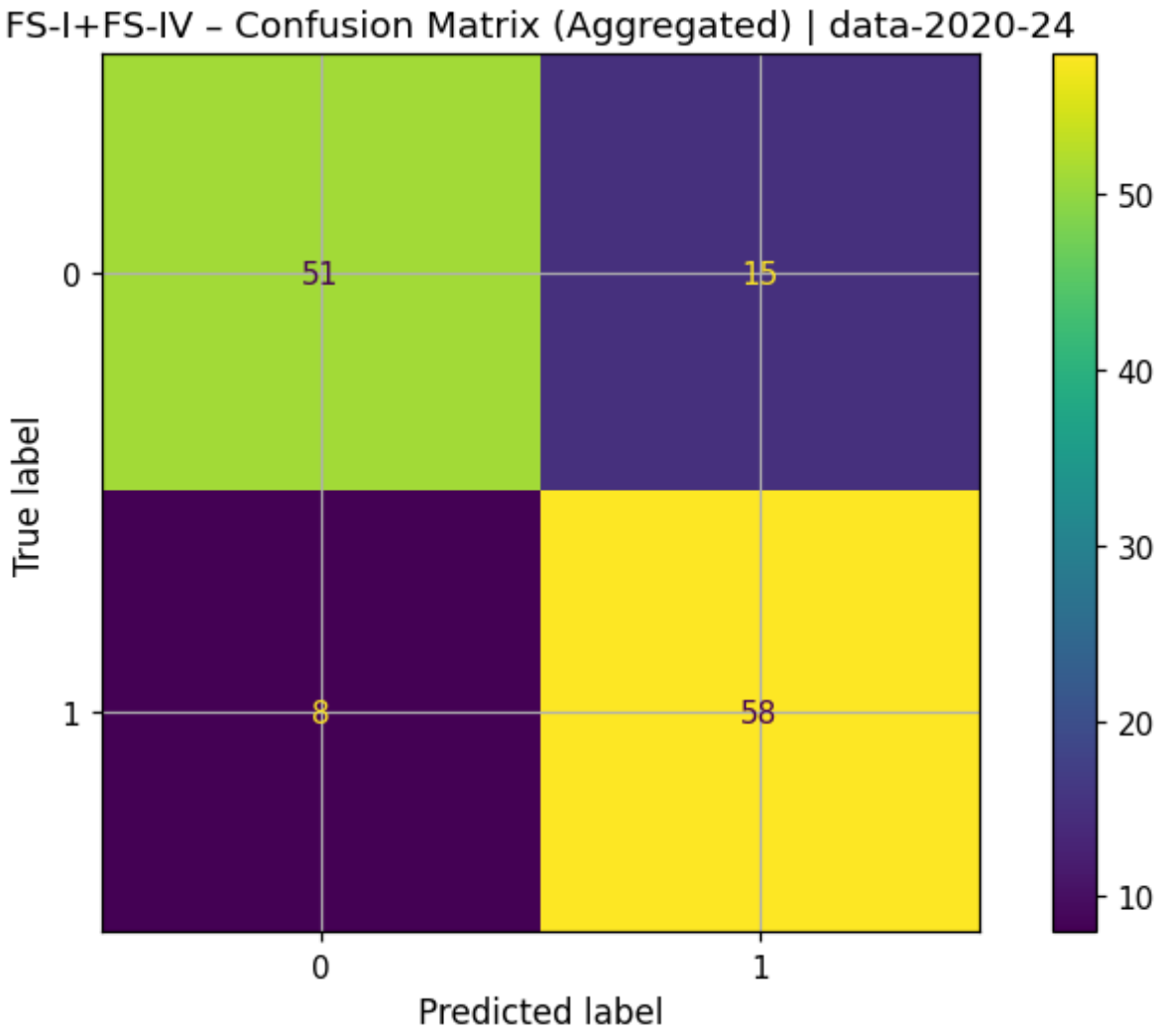


Figure 7: 2020–2024 — Confusion Matrix

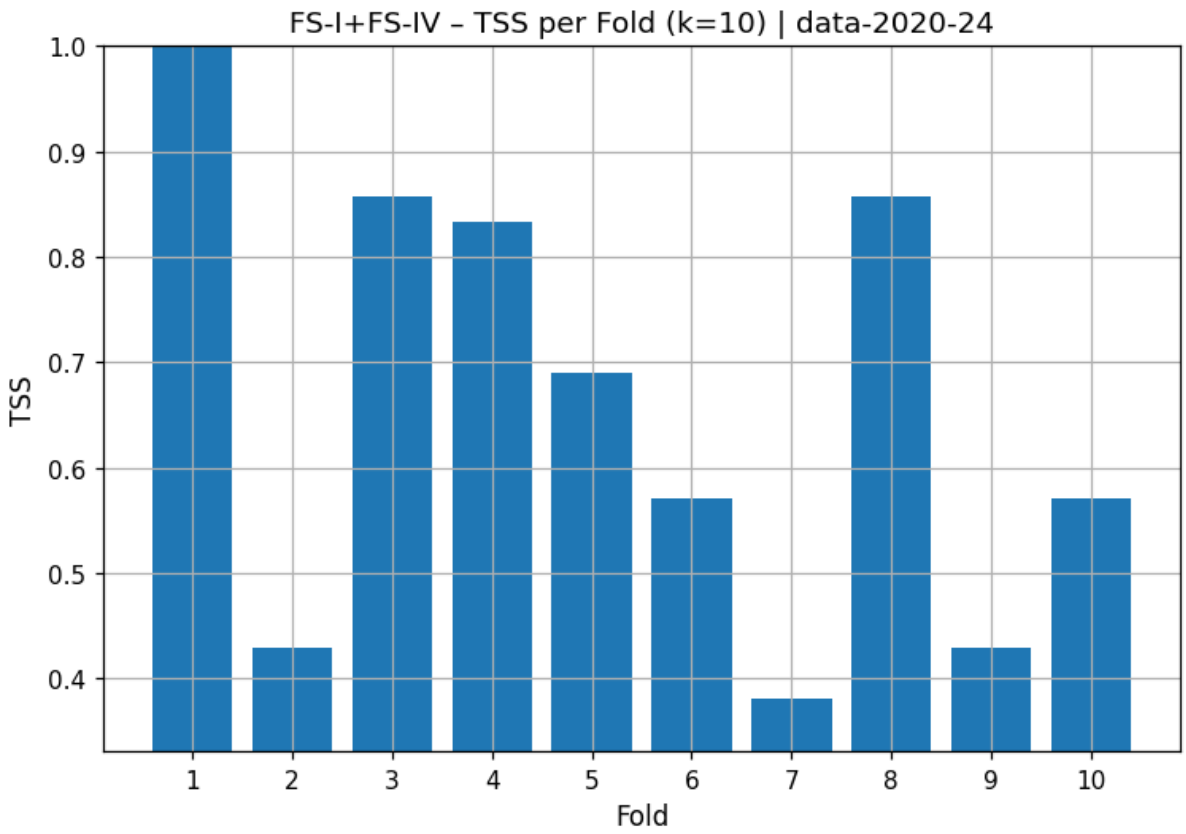


Figure 8: 2020–2024 — TSS per Fold

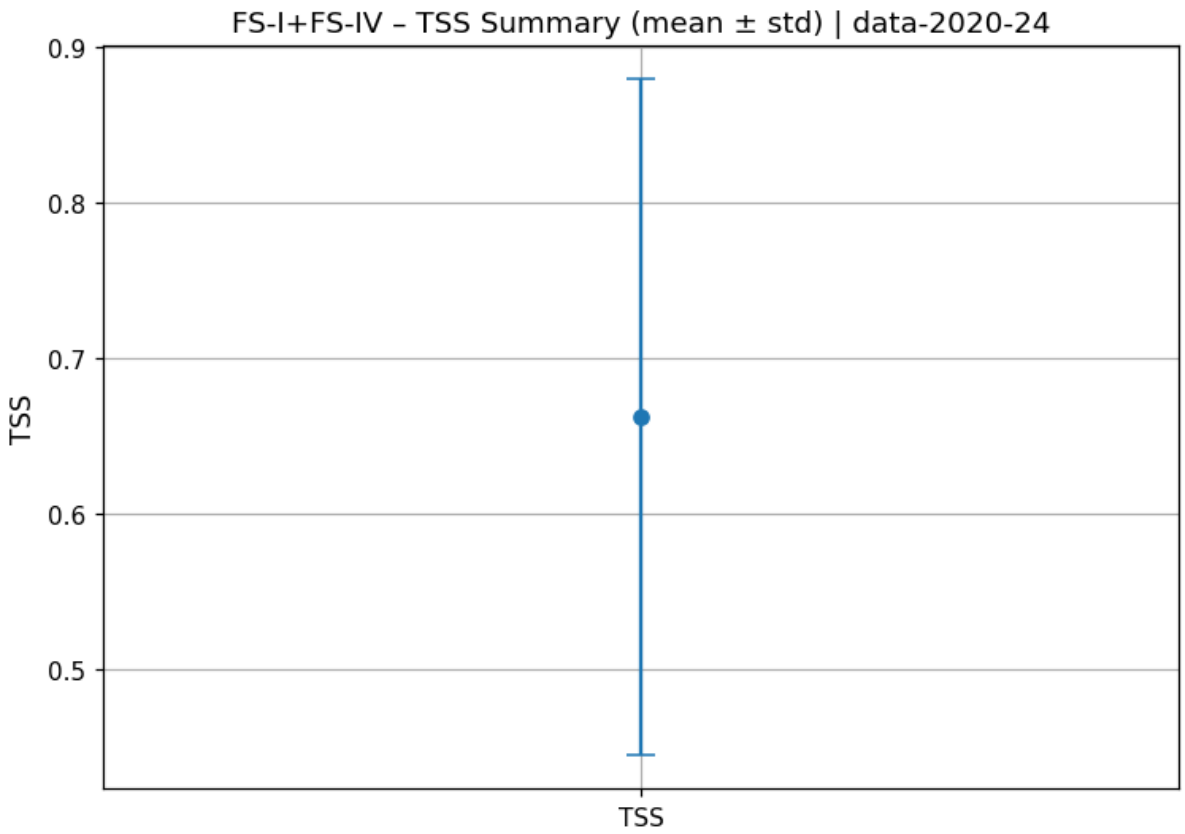


Figure 9: 2020–2024 — TSS Summary (mean ± std)

3. Discussion / Short Answers

(a) Which feature combination worked best and which was worst?

Best: The combination **FS-I + FS-IV** achieved the highest overall performance with a True Skill Statistic (TSS) of 0.7773 ± 0.0395 on the 2010–2015 dataset, demonstrating that combining the main magnetic features (FS-I) with the max–min range features (FS-IV) yields the most discriminative representation for flare prediction.

Worst: The **FS-III** feature set (Historical Activity only) produced the lowest performance, with a TSS of 0.0000 ± 0.0000 over $k = 10$ folds. This indicates that past flare counts alone are not predictive of future events, as the model effectively performs no better than random guessing.

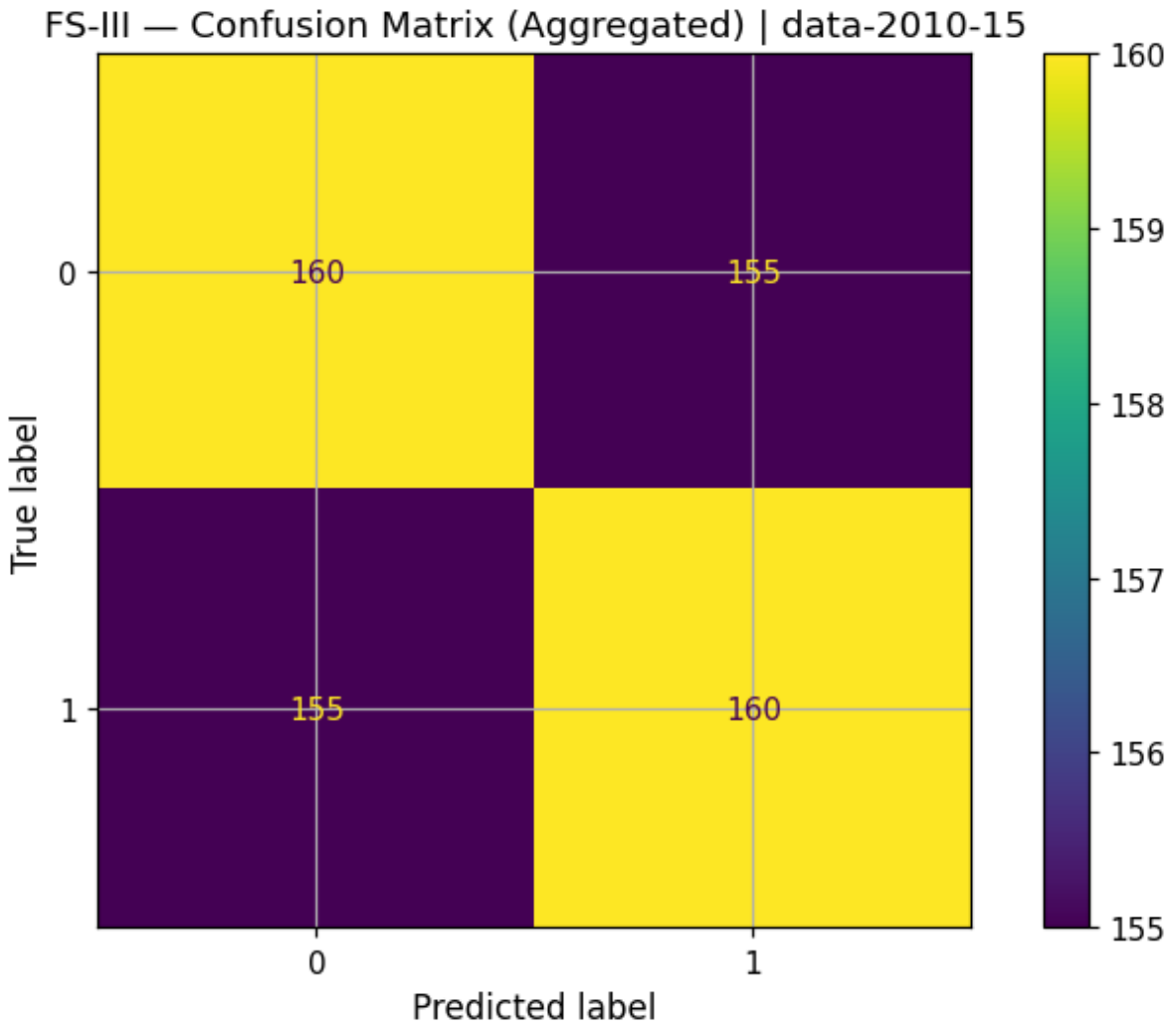


Figure 10: FS-III — Confusion Matrix and TSS Summary (mean ± std = 0.0000 ± 0.0000).

(b) Does adding additional feature sets improve TSS?

Yes. Combining FS-I (main features) with FS-IV (max–min features) improved TSS by incorporating both magnitude and range information, enhancing discrimination between flare and non-flare regions.

(c) Which dataset led to a better TSS and why?

The 2010–2015 dataset outperformed the 2020–2024 dataset (TSS 0.777 vs. 0.662). This difference likely stems from greater data consistency, lower measurement noise, and more balanced event representation in the 2010–2015 dataset.